Condensation Dynamics

Group 5: Aditya Prasad (22110018), Chandra Shekhar (22110056), Hansin Shah (22110090), Hari Balaji (22110092), Manas Kalal (22110138), Mayank Dahotre (22110147), Pranav Kamboj (22110198), Srirangan S. (22110259), Dev Vyas (24120005)

Department of Mechanical Engineering

Prof. Soumyadip Sett

1. Introduction

Condensation occurs when water in the water vapour in the air transforms into liquid droplets upon contact with a cooler surface. This project investigates the condensation dynamics on different metal surfaces of different types. By manually lowering the temperature of the metal surface below the dew point and inducing condensation, this project studies the rate of condensation on material and surface area.

2. Problem Statement

The goal of this study, "Condensation Dynamics on Metal Plates," is to systematically examine the rates of condensation on different surfaces and materials. Using a thermoelectric Peltier cooling device, controlled thermal conditions are used to study smooth and finned copper and aluminum surfaces. By examining the dynamics of condensation on copper and aluminum surfaces with smooth and finned geometries, In order to cause condensation, the study will gradually reduce surface temperatures below the dew point using a thermoelectric Peltier cooler. In a variety of controlled thermal and material circumstances, the study attempts to measure key metrics such as the rate of condensation, water collecting mass, and condensation heat flux. It is anticipated that the study's conclusions will contribute to our basic knowledge of condensation processes. The experimental setup will be designed and developed to study condensation dynamics. As the surface temperature falls below the dew point temperature, atmospheric vapor condenses on the surfaces. The condensation rate can be controlled by controlling the Peltier temperature connected to the DC voltage source. The higher the temperature difference between the ambient and the Peltier, the cooler the surface; thus, it will have a higher rate of condensation. The rate of condensation can be quantified from the measured water collected off the surface. So, condensation heat flux for different condensation rates on different surfaces can be compared at the end.

3. Governing Equations and other parameters

Condensation on a Vertical Surface: Derivation

Governing Principles

Nusselt's theory for laminar film condensation is the most fundamental theory that governs the film wise condensation.

Force Balance

For a liquid film flowing down a vertical plate:

$$\tau\, dx = (\rho - \rho_v)(\delta - y)\, g\, dx$$

Using **Newton's law of viscosity**, where the shear stress is given by $\tau = \mu \frac{du}{dy}$, this becomes:

$$\mu \frac{du}{dy} dx = (\rho - \rho_v)(\delta - y) g dx$$

Boundary conditions:

$$u = 0 \quad \text{at} \quad y = 0$$

Integrating, the velocity profile is:

$$u(y) = \frac{g(\rho - \rho_v)}{\mu}\left(\delta y - \frac{y^2}{2}\right)$$

where $u(y)$ is the velocity at a point y in the film of thickness δ.

Mass Flow Rate

The mass flow rate $m(x)$ per unit width of the plate is given by:

$$m(x) = \int_0^\delta \rho u\, dy$$

Substitute $u(y)$:

$$m(x) = \int_0^\delta \rho \frac{g(\rho - \rho_v)}{\mu}\left(\delta y - \frac{y^2}{2}\right) dy$$

Evaluate the integral:

$$m(x) = \frac{\rho g(\rho - \rho_v)}{\mu}\left[\frac{\delta^3}{2} - \frac{\delta^3}{6}\right]$$

$$m(x) = \frac{\rho g(\rho - \rho_v)\delta^3}{3\mu}$$

Heat Transfer Balance

The rate of heat transfer by conduction through the film is:

$$dQ_{\text{conduct}} = k \frac{T_v - T_w}{\delta} dx$$

The heat transfer due to condensation of vapor is:

$$dQ_{\text{condens}} = \lambda\, dm$$

Equating the two:

$$\lambda\, dm = k \frac{T_v - T_w}{\delta} dx$$

Substitute $dm = \frac{\partial m}{\partial \delta} d\delta$:

$$\frac{\partial m}{\partial \delta} d\delta = \frac{k(T_v - T_w)}{\lambda \delta} dx$$

Film Thickness

From the mass flow rate derivative:

$$\frac{\partial m}{\partial \delta} = \frac{\rho g(\rho - \rho_v)\delta^2}{\mu}$$

Substitute this into the heat transfer equation:

$$\frac{\rho g(\rho - \rho_v)\delta^2}{\mu} d\delta = \frac{k(T_v - T_w)}{\lambda \delta} dx$$

Rearranging:

$$d\delta = \frac{\mu k(T_v - T_w)}{\rho g(\rho - \rho_v)\lambda \delta^3} dx$$

Integrating, with $\delta = 0$ at $x = 0$:

$$\delta(x) = \left[\frac{4\mu k(T_v - T_w)x}{\rho g(\rho - \rho_v)\lambda}\right]^{1/4}$$

Local Heat Transfer Coefficient

The local heat transfer coefficient h_x is:

$$h_x = \frac{k}{\delta(x)}$$

Substitute $\delta(x)$:

$$h_x = \left[\frac{g\rho(\rho-\rho_v)\lambda k^3}{4\mu(T_v-T_w)x}\right]^{1/4}$$

Average Heat Transfer Coefficient

The average heat transfer coefficient over a length L is:

$$h_m = \frac{1}{L}\int_0^L h_x dx = \frac{4}{3}h_x\Big|_{x=L}$$

Substitute h_x at $x = L$:

$$h_m = 0.943\left[\frac{g\rho(\rho-\rho_v)\lambda k^3}{\mu(T_v-T_w)L}\right]^{1/4}$$

3.1. Assumptions of Nusselt Theory

1. Steady-State Conditions: The liquid film reaches a stable thickness over time.
2. Laminar Flow: The condensate film remains laminar throughout.
3. Negligible Inertia and Shear Stress: The effects of inertia and shear stress at the film-vapor interface are negligible.
4. Constant Properties: Thermophysical properties of the liquid are assumed constant.
5. Thin Film Approximation: The film thickness is much smaller than the characteristic dimensions of the surface.

3.2. Special Note on Steady-State Assumption

In our experiment, confirming the attainment of steady-state conditions is inherently challenging due to the dynamic local conditions.

Furthermore, variations in surface temperature, humidity levels, and local flow dynamics introduce transient behavior, making it unlikely that steady-state is achieved within the experimental timeframe. Despite these complexities, the classical Nusselt theory for film condensation provides a fundamental framework for understanding the process. Deviations between the experimental results and theoretical predictions may be attributed to the transient nature of the process and the . This emphasizes the importance of considering such factors in analyzing and interpreting the data.

4. Experimental Setup

4.1. Components

1. **Thermoelectric Peltier Cooler:** The Thermoelectric Peltier Cooler is a cooling device that utilises the Peltier effects to transfer heat from one side of the device to the other when electric current is applied, thereby cooling one side while simultaneously heating the other. By cooling the surface of the test sample below the dew point, it triggers condensation.
2. **Aluminium Heat Sink and fan:** The purpose of the heat sink is to help dissipate heat effectively by facilitating strong heat conduction. By dissipating heat from the hot side of the Peltier cooler, it ensures that a lower temperature is maintained on its cold side, which is crucial for condensation.
   The fan provides circulation of air over the fins of the heatsink so that the heat from the sink is radiated to the environment more effectively.
3. **Humidifier:** The humidifier increases the moisture content in the air, when kept next to the setup and provides a more ideal environment for condensation, ensuring there is enough water vapour continuously available for condensation.
4. **Test Samples:** The condensation rates of four samples of two materials (Copper and Aluminium with and without fins) are to be compared. This can help understand the dependence of condensation on the materials, as well as the surfaces.
5. **Thermocouple:** The Thermocouple measures the surface temperature at any time, monitoring and controlling the conditions for optimal condensation. It is required to make sure the surface temperature drops below the dew point.
6. **Petri dish:** A petri dish used to collect the condensed droplets of water from the surface of the sample. By measuring the mass of the water in the dish, the rate of condensation can be observed experimentally.
7. **Variable DC Voltage Source:** A variable DC Voltage source is required to power the Thermoelectric Peltier Cooler and ensure the heat transfer is occurring continuously.
8. **Fixed voltage DC source:** This was used to power the 4 mm DC fans for generating the air current for the novelty of this experiment.
9. **4 mm DC fans (24 V):** These were used to blow the air from the humidifier vent onto the surface of the Peltier cooler to demonstrate the effect of airflow on the results of condensation, which was the novelty of this experiment.
10. **Thermal Compound:** This is a paste used to increase contact between the heat sink and the conducting surface so that heat transfer occurs effectively.
11. **MDF and Acrylic panels:**
    - These panels allow all the components to be integrated together seamlessly.
    - The MDF panels form the main structure of the setup.
    - The base MDF panel has a cutout for the humidifier so that its vapour vent is inside the enclosure.
    - The acrylic panels allow an enclosure to be created so that the rate of escape of the humid air is very low, and also allow a clean view of the experimental process.
    - The top acrylic panel is removable and also has a hole in it for the thermocouple wire.
12. **Extension Cord:** This is used to connect the DC Voltage source and the DC fan to a power socket in the wall.

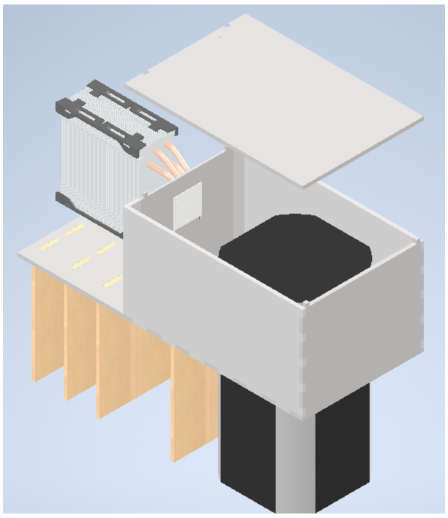

Figure 1. CAD Model of the experimental Setup

The Thermoelectric Peltier Cooler is mounted onto a heat sink, maintaining proper thermal contact between the two.
The DC Fan is fixed close to the other side of the heat sink to further improve heat dissipation from the hot side of the Peltier cooler.
The test sample is attached on the other (cold) side of the cooler, once again carefully ensuring proper thermal contact.

The Peltier cooler, via connecting wires is powered with the variable DC voltage source.
The probe of the Thermocouple is kept in contact with the surface of the test sample, and actively measures the surface temperature.
To elevate the assembly, a fixture/stand is made with MDF sheets, that keeps the cooler, heat sink and sample secured tightly in a higher altitude.
To collect the condensed water droplets, a beaker is positioned under the test sample.
The humidifier is placed close to the setup to maintain the humidity level.

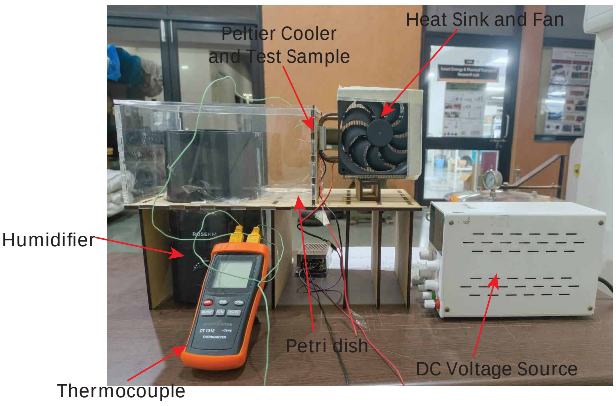

Figure 2. Completed Experimental Setup

5. Experimental Methodology

The following outlines the step-by-step process for conducting experiment to analyze the condensation dynamics phenomenon on metal plates with varying materials of same surface geometry.

5.1. Preparation of Experimental Setup

1. We assembled the experimental setup as described in the *Experimental Setup* section.
2. We mounted the Thermoelectric Peltier Cooler onto the heat sink using thermal compound to ensure good thermal contact.
3. Also, we attached the test sample (smooth or finned copper/aluminum plate) on the cold side of the Peltier cooler using thermal compound for efficient heat transfer.
4. Further, we secured the entire assembly on the MDF stand and position the acrylic panels to create an enclosure.
5. Afterwards, we placed the humidifier near the enclosure with its vent directed into the setup to ensure a humid environment.
6. Finally, we positioned a petri dish directly under the test sample to collect condensed water droplets.

5.2. Condensation Experiment

1. We calibrate the thermocouple by comparing its readings to a known standard to ensure accurate temperature measurements.
2. Next, we verified the functionality of the DC voltage source, fan, and humidifier.
3. Later, we turned on the humidifier and allow the enclosure to fill with moist air.
4. Afterwards, we measured and recorded the ambient temperature, relative humidity, and dew point temperature using a thermocouple. *Note: dew point temperature is approximately 10 degrees C less than ambient temperature*
5. Furthermore, we powered the Peltier cooler using the constant DC voltage source, for each reading for maintaining a surface temperature of the test sample below the dew point.
6. Soon, we monitored the surface temperature of the test sample using the thermocouple.
7. Also we observed the initiation of condensation as water droplets form on the sample surface.
8. Lastly, we collected condensed water droplets in the petri dish positioned below the sample.

5.3. Data Collection

For each sample (smooth and finned copper and aluminum), we performed the following steps:

1. We maintained the surface temperature at a specific value below the dew point (e.g., $-5°C$, $-10°C$).
2. Also, we measured and record the surface temperature at regular intervals after a set duration, and weighed the collected water in the petri dish using a precision balance to determine the rate of condensation.
3. Next, We repeated the process for different temperature settings to analyze the effect of surface temperature on condensation rates.
4. This process was repeated for around 65 hours straight over 13 intervals of 5 hours
5. For each interval, there was a different ambient temperature depending on the time of day. So the ambient temperature for each interval was considered and the calculations for that interval were made accordingly.
6. Mainly, we recorded Voltage, Surface temperature of the test sample, ambient temperature and the constant Voltage applied
7. Furthermore, we repeated the experiment for each of the four samples (smooth copper, finned copper, smooth aluminum, finned aluminum).
8. Simultaneously, we recorded all observations, weights, temperatures, and calculated heat flux values in a systematic format (a spreadsheet, in our case).
9. These records were then used to interpret results and draw conclusions about condensation dynamics on the different surfaces.

By following this methodology, the experiment yielded quantitative insights into the condensation processes influenced by material properties, surface geometry, and thermal conditions.

6. Results

The experiments were performed for 3 different surfaces - a flat Aluminium plate, a flat Copper plate and an Aluminium plate with extended surfaces, each with dimensions of 4 cm x 4 cm. And for each of the samples, the readings were taken setting the surfaces to 3 unique temperatures.

6.1. Mass Collected

The mass collected in each cycle of 5 hours were recorded and the 9 readings taken are represented as follows:

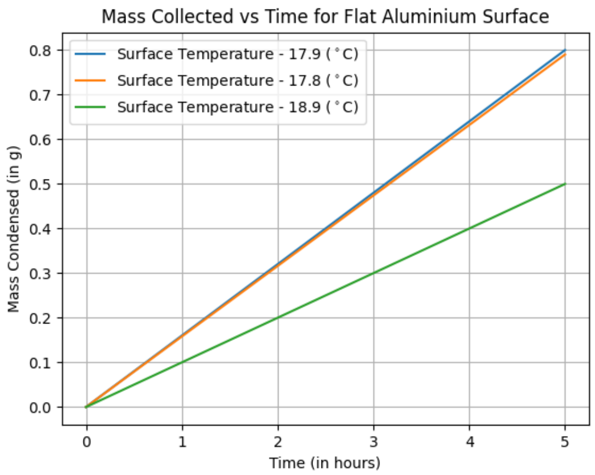

Figure 3. Mass Collected Over Time for Flat Aluminium Surface

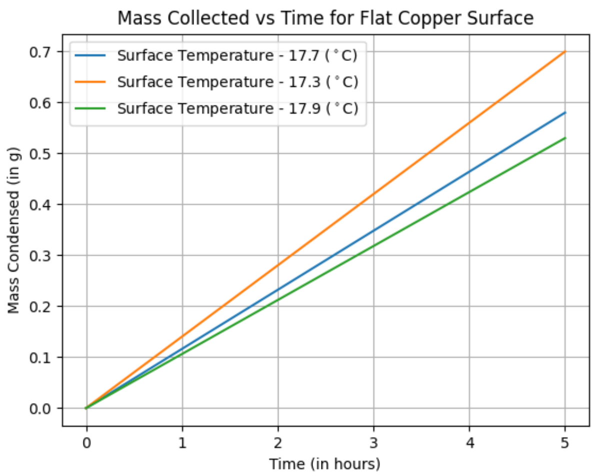

Figure 4. Mass Collected Over Time for Flat Copper Surface

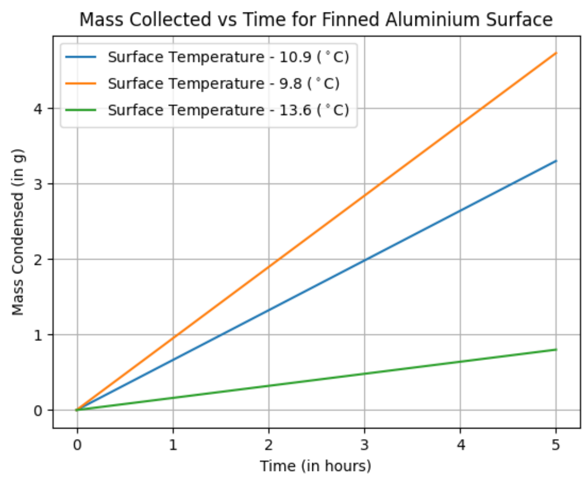

Figure 5. Mass Collected Over Time for Finned Aluminium Surface

Alternatively, the 3 different surfaces can be compared with one another by plotting the corresponding curves in the same graph. However, since a uniform surface temperature could not be maintained across all 3 surfaces, the condensation occurring when the same amount of voltage is supplied to the Peltier cooler is compared.

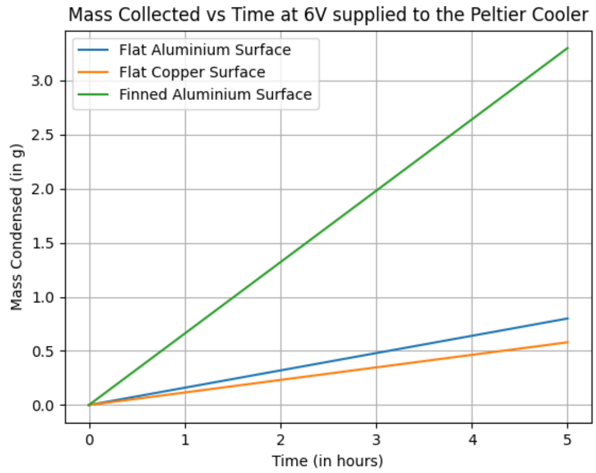

Figure 6. Mass Collected Over Time at 6V supplied to the Peltier Cooler

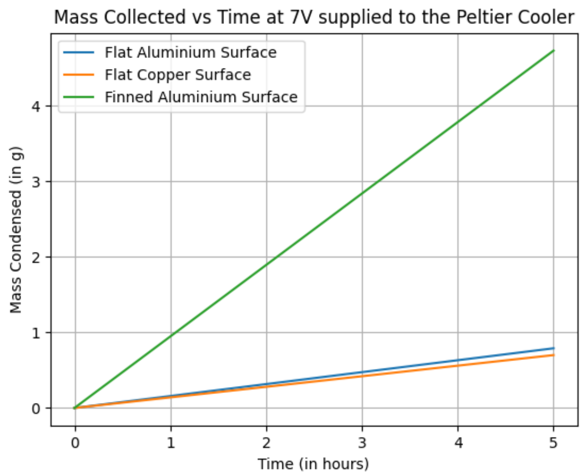

Figure 7. Mass Collected Over Time at 7V supplied to the Peltier Cooler

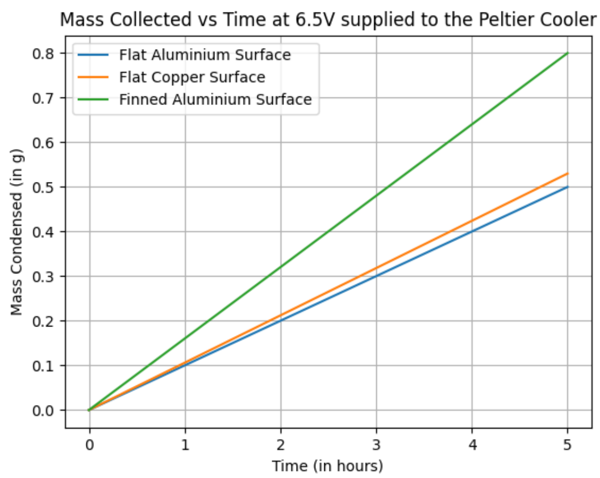

Figure 8. Mass Collected Over Time at 6.5V supplied to the Peltier Cooler

6.2. Condensation Heat Flux

The condensation heat flux is calculated assuming a latent heat of condensation of 2260 kJ/kg. The obtained results are tabulated below:

Surface	Surface Temperature (°C)	Mass Collected (g)	Condensation Rate (g/(hr.m^2))	Condensation Heat Flux (W/m^2)
Flat Aluminium	17.9	0.79	100.00	62.78
Flat Aluminium	17.8	0.80	98.75	61.99
Flat Aluminium	18.9	0.50	62.50	39.24
Flat Copper	17.7	0.58	72.50	45.51
Flat Copper	17.3	0.70	87.50	54.93
Flat Copper	17.9	0.53	66.25	41.59
Finned Aluminium	10.9	3.30	412.50	258.96
Finned Aluminium	9.8	4.73	591.25	371.17
Finned Aluminium	13.6	0.80	100.00	62.78

Table 1. Condensation Data for Different Surfaces

6.3. Condensation Rate and Temperature

For each of the surfaces, the condensation rate is plotted as a function of the surface temperature.

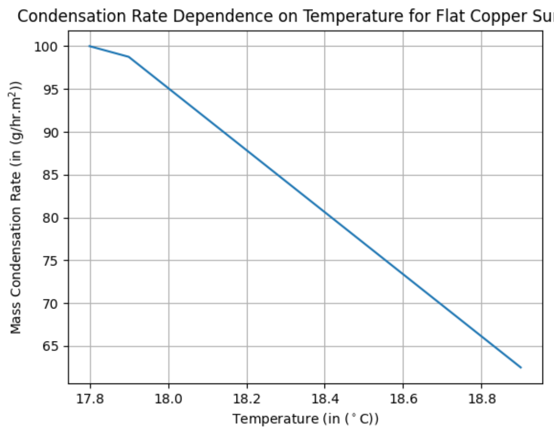

Figure 9. Condensation Rate versus Temperature for Flat Aluminium Surface

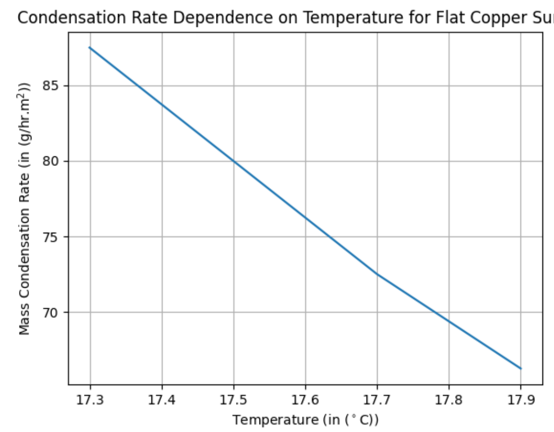

Figure 10. Condensation Rate versus Temperature for Flat Copper Surface

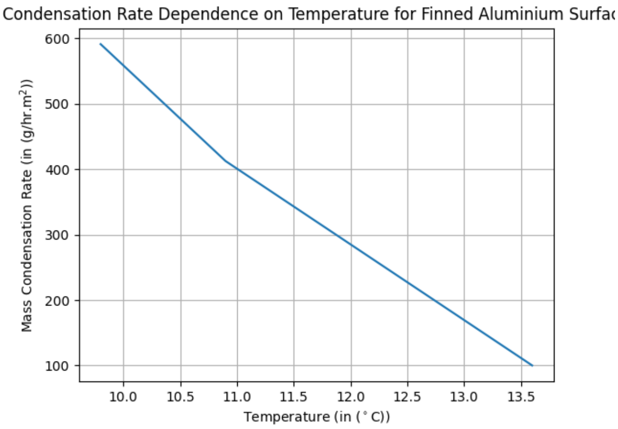

Figure 11. Condensation Rate versus Temperature for Finned Aluminium Surface

7. Discussions and Inferences

7.1. Mass Collected

Within each of the individual samples except for the finned Aluminium surface, it can be observed that the mass condensed in the same period of time is not varying much with the surface temperature. However, in the case of the finned Aluminium surface, the amount of water condensed is significantly lower when the surface temperature is higher. This could be argued that while the surface temperature in the two flat surfaces did not drop down much and remained over the same range at different voltages supplied, there was significant drop in the surface temperatures in the case of the finned Aluminium surface and hence a higher difference in temperatures and thus a larger variation in the mass of water condensed.
From the graphs obtaining comparing each surface's performance at a given voltage supplied to the Peltier cooler, it can inferred that the finned surface provides much better condensation than the other two surfaces. The two flat surfaces (Cu and Al), in each of the cases, condensed a similar mass of water.

7.2. Condensation Heat Flux

The condensation heat flux is ideally calculated using Nusselt theory and is given by

$$q'' = h_c(T_s - T_{\text{sat}}) \tag{1}$$

where

$$h_c = \frac{k_l}{\delta} \tag{2}$$

- k_l is the conduction coefficient at liquid state (of the film).
- δ is the film thickness.

However, since no proper film was formed on the test surface, the condensation heat flux is alternatively calculated as

$$q'' = \dot{m}h_f \tag{3}$$

where h_f is the latent heat of condensation and

$$\dot{m} = \frac{m}{A_s \cdot t} \tag{4}$$

- $\dot{m}$ is the mass flux of the condensate.
- m is the mass of water condensed over the given period of time.
- A_s is the surface area of condensation.
- t is the time taken into observations of condensation.

The latent heat of condensation is taken using the known values available for given compound at given temperature.
The finned surfaces are observed to have higher condensation heat fluxes. Additionally, comparing within the same surface, a higher

condensation heat flux is observed at a lower surface temperature. This aligns with the mass collected and is verified analytically as the condensation heat flux is directly proportional to the mass collected.

7.3. Temperature Dependence on Condensation Rate

From the graphs obtained, it can be seen that lower the surface temperature that is achieved, higher the rate of condensation. Among all the samples, the highest condensation rate is observed at the lowest surface temperature. The experimental results show that even though the graph is not definitively linear with a negative slope, higher the surface temperature, the lesser the condensation that will occur.

7.4. Benchmark Experiments

With the condensation occurring at a sealed environment, and the humidifier releasing humid air as well, there was the possibility of the petri dish collecting the microdroplets from the humid air along with the condensed water from the surface. Hence, to identify and negate the possible error, the humidifier was run in the environment in the absence of surface cooling, and the mass of collected humid microdroplets were measured in the same time frame. Consequently, while noting down the final reading, the mass of condensed water was evaluated as the difference between mass of water collected and the measured mass effect of the humidifier.

7.5. Film Formation

Nusselt's Theory states that a film originating from the top of the plate flowing downwards is formed during condensation. This occurs when the condensation process has reached steady state and condensation rate, heat transferred through the film and all other properties are independent of time. A few possible reasons for the film not forming are:

1. Slight contamination/scratches on the test surface might resist the formation of a continuous film.
2. It is required to achieve steady state before observing a proper film formed on the surface.
3. The experiment was performed in a humid environment supplied by a humidifier. The humidifier air velocity could have caused the drops to shear before becoming a continuous film.
4. The amount of water condensed was too small for a proper film to form and be visible on the surface.

7.6. Reasons for Discrepancy

- The experiments for different surfaces were performed at different humidifier speeds. While that affects neither the surface properties nor the ambient temperature, there is an excess of water vapour content in the ambient air, providing more vapour for condensation.
- The surface temperature was controlled by adjusting the voltage of the DC Voltage Source powering the Peltier cooler. The surface temperature was mildly fluctuating throughout the experiment. To maintain uniformity throughout the readings, it was not possible to accordingly regulate the DC Voltage output to get uniform surface temperature across surfaces. Additionally, having a higher voltage input did not necessarily provide a lower surface temperature as the cold side of the Peltier cooler began to heat up beyond a certain voltage.
- A very small percentage of the condensed droplets trickled down the flat surfaces to the supporting stand. When there were extended surfaces, this problem did not occur.
- The experiment was to be conducted for 4 different surfaces. However, since only a 1.3 cm x 1.4 cm sized Copper extended surface was procured, the experiment was not performed for that surface. An option considered was to attach the small Copper sample. However, since that would only cover one-ninth of the Peltier cooler area, water would also condense off the surface of the cooler, thereby staggering the results.
- A more accurate representation of the mass collected curve would have been possible had the mass readings been taken periodically. However, the petri dish having to be removed in between opened the chance of the droplets falling immediately during the few seconds of removal. Blowing down the droplets on the test surface was not an option as that could disturb the potential formation of a film.

8. Conclusions

By building an experimental setup and performing practical analyses over a large time, the following conclusions were reached.

- Condensation occurs in sparse amounts, even in closed humidity supplied conditions, evidenced as not more than 5 grams of water was condensed in any of the surfaces at any surface temperature.
- Among the three surfaces tested, the Aluminium plate with extended surfaces (fins) showed to be significantly better at condensation due to its increased surface area available for heat transfer.
- Subsequently, the highest condensation rate and the highest condensation heat flux were shown by the Aluminium finned surface.
- Both the flat surfaces - Copper and Aluminium showed similar rates of condensation which was well below the corresponding value for the finned Aluminium plate.
- Lower surface temperatures produced a higher condensation heat flux in all of the materials.
- No formation of continuous film was observed during the experiment.

9. Novelty

We decided to investigate the effect of airflow on the condensation rate to see what would happen. By placing a set of small fans near the vent of the humidifier, we were able to generate an air current towards the peltier, and then observe the effects of condensation on its surface. To power the fans, a fixed voltage DC source was used.

Figure 12. The humidifier with the fans near the vents

9.1. Results

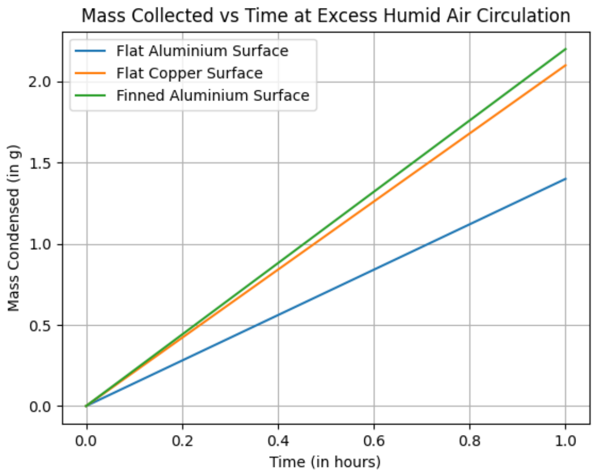

Figure 13. Mass Collected in Time at Excess Humidity

Surface	Surface Temperature (°C)	Mass Collected (g)	Condensation Rate (g/(hr.m²))	Condensation Heat Flux (W/m²)
Flat Aluminium	14.9	1.4	875	549.31
Flat Copper	14.6	2.1	1312.50	823.96
Finned Aluminium	11.4	2.2	1375	863.19

Table 2. Condensation Data for Different Surfaces at Excess Humidity

9.2. Discussions

When an excess air current was generated close to the peltier, and humid water vapour droplets in the vicinity of the surface were present, a huge increase in the mass collected, condensation rate and other parameters was observed. While it could be partly credited to the humidifier due, for which benchmark experiments performed give an idea on how much it contributes to the mass collected, there is also an increase in the condensation rate significantly, as adding the humid air current close to the surface made it easier for the surface to condense.

The extra condensation is due to the incidence of the humid air current on the peltier cooler. The condensation due to the cooling effect along with the condensation due to the air current contributes to the total amount of water condensed on the cooler, and it explains why a higher mass of water was collected in this case compared to running the experiment without the fans.

10. Critical Experimental Analysis

The project's main goal of examining the dynamics of condensation in a range of materials and geometries was accomplished. Nevertheless, several restrictions have been noted:

1. Experimental Limitations: The studies were carried out in environments with varying humidity and ambient conditions, which resulted in inconsistent findings.
2. Surface-Related Challenges: Minor surface flaws, temporary circumstances, and disruptions from external airflow produced by the humidifier were the main reasons why the expected development of continuous condensate films was not achieved.
3. Problems with Voltage Control: The Peltier cooler's performance varied at high voltages, which occasionally reduced cooling effectiveness. This fluctuation made it more difficult to keep the surface temperature constant during the experiment.
4. Accuracy of Data Collection: On surfaces with high rates of condensation, the intermittent removal of the petri dish for mass measurements occasionally interfered with the condensation process, which could result in errors.

10.1. Challenges Faced

1. Environmental Variability: Attempts to create consistent experimental conditions were hampered by changes in the surrounding temperature and humidity over time.
2. Unsturdy Setup Prototype: The Setup components of the prototype were placed in an ad-hoc manner, which hampered our initial experimental efforts.
3. Instrumental Limitations: The voltage source and thermocouple were functional but lacked the exact control required for exacting experimental regulation.
4. Human error: Manual modifications to the voltage settings and periodic handling of the apparatus led to unavoidable variations in the experimental process.
5. Lack of samples: We could not obtain enough of the finned copper surfaces to completely cover the surface of the Peltier cooler, and hence we could not obtain readings for that set of samples.

11. Future Scope of Work

1. Improved Experimental Design:
   - Establish a sealed and climate-controlled chamber to standardize humidity and temperature conditions throughout the experimental phases.
   - Implement automated systems to accurately control and maintain the surface temperature of the Peltier cooler.
   - Integrate advanced sensors for real-time monitoring of droplet size, condensation patterns, and film formation.
2. Energy Efficiency and System Optimization:
   - Prioritize improving Peltier cooler efficiency by exploring alternative configurations or cooling mechanisms.
   - Assess the trade-offs between power consumption and condensation performance to achieve optimal energy efficiency.
3. Potential Innovations:
   - Investigate passive cooling systems that function based on natural temperature gradients for condensation, thereby minimizing reliance on powered devices.
   - Explore hybrid systems that integrate Peltier cooling with alternative methodologies, such as evaporative cooling or phase-change materials.

By addressing these areas, future research can effectively bridge the gap between experimental findings and practical applications, thereby enhancing the robustness and impact of the study on condensation dynamics.

12. Contributions

1. **Aditya Prasad (22110018):**
   - Helped in choosing the required experiment components
   - Contributed to the Phase 1 report by writing the Essential Background section.
   - Contributed to the efforts to make the setup more sturdy before starting the readings
   - Took experimental readings in a 5-hour shift
   - Wrote the 'Critical Analysis' and 'Future Scope of Work' sections in the Phase 3 report

2. **Chandra Shekhar (22110056):**
   - Component and equipment selection
   - Wrote sections on governing equations and essential parameters (phase 1 report).
   - Verified the functionality of the DC voltage source and fan.
   - Collected preliminary experimental readings.
   - Helped assembling the experimental setup.
   - Provided inputs in designing the experimental setup.
   - Recorded experimental readings in extended five-hour shifts.
   - Drafted the "Results" section of the final project report.
3. **Hansin Shah (22110090):**
   - Acquisition of material and equipment.
   - Contributed in the design of the prototype.
   - Sat one 5 hour shift to take experimental readings for phase 3
   - Wrote the Critical Analysis and Future scope of work sections in Phase 3 report
4. **Hari Balaji (22110092):**
   - Material and equipment acquisition.
   - Initial numerical calculations for the experiment (phase 1).
   - Design of the prototype (phase 2).
   - Initial testing of the prototype functionality (phase 2).
   - Video recording and editing for phase 2.
   - CAD modeling, design and assembly of final setup (phase 3).
   - One shift of readings (phase 3).
   - Detailed description of the experimental setup and methodology (phase 3 report).
5. **Manas Kalal (22110138):**
   - Assisted in component selection.
   - Drafted the section on governing equations and analyzing other essential parameters.
   - Helped with laser cutting to make the parts for the experiment setup.
   - Collected experimental readings in a preliminary 1-hour and final 5-hour shifts.
   - Effectively showcased the setup and results in the phase 2 video.
   - Contributed to writing the Results section in the final report.
6. **Mayank Dahotre (22110147):**
   - Assisted in the selection of components required for the experimental setup
   - Drafted the section on Experimental Setup and Methodology in the phase 1 report.
   - Took preliminary readings of the condensation (phase 2)
   - Contributed in assembly of the final experimental setup and made initial verifications
   - Supervised the Methodology Section (Phase 3 Report).
7. **Pranav Kamboj (22110198):**
   - Acquisition of material and equipment.
   - Initial numerical calculations for the experiment for phase 1.
   - Design of the prototype for phase 2.
   - Initial testing of the prototype for functionality for phase 2, along with initial readings.
   - Video recording and editing for phase 2.
   - One shift of readings for phase 3.
   - Detailed description of the experimental setup and methodology for phase 3.
8. **Srirangan S. (22110259):**
   - Detailed the proposed experimental setup (phase 1 report).
   - Laser cut initial setup structure.
   - Took preliminary readings of condensation (phase 2).
   - Explained the experiment in the video.
   - Fabricated and assembled final experimental setup.
   - Verified the functionality of all the components.
   - One shift of condensation readings.
   - Coordinated with the project mentor whenever necessary.
   - Detailed the discussions section of the final phase-3 report.
9. **Dev Vyas (24120005):**
   - Component selection.
   - Drafted the section of "Proposed Analysis" (phase 1 report).
   - Verified the functionality of the DC voltage source and fan.
   - Helped assemble preliminary setup.
   - Took preliminary condensation readings (phase 2).
   - Provided inputs to improve the experimental setup (phase 3).
   - Took condensation readings during a 5-hour shift.
   - Drafted the "Governing Equations" section of the final project report along with assumptions.

References

[1] F. P. Incropera and D. P. DeWitt, *Fundamentals of Heat and Mass Transfer*, 4th. John Wiley & Sons, 1996, ISBN: 978-0471304609.